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MINI-REVIEW ARTICLE

Structure-Bioactivity Relationship Study of Xanthene Derivatives: A Brief Review

Aref G. Ghahsare¹, Zahra S. Nazifi^{1,*} and Seyed M.R. Nazifi^{2,*}

¹Department of Chemistry, Shahreza Branch, Islamic Azad University, Shahreza, Isfahan, 86145-311, Iran; ²Department of Medicinal Chemistry, School of Pharmacy and Pharmaceutical Sciences, Isfahan University of Medical Sciences, 81746-73461 Isfahan, Iran

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Abstract: Over the last decades, several heterocyclic derivatives compounds have been synthesized or extracted from natural resources and have been tested for their pharmaceutical activities. Xanthene is one of these heterocyclic derivatives. These compounds consist of an oxygen-containing central heterocyclic structure with two more cyclic structures fused to the central cyclic compound. It has been shown that xanthene derivatives are bioactive compounds with diverse activities such as anti-bacterial, anti-fungal, anti-cancer, and anti-inflammatory as well as therapeutic effects on diabetes and Alzheimer. The anti-cancer activity of such compounds has been one of the main research fields in pharmaceutical chemistry. Due to this diverse biological activity, xanthene core derivatives are still an attractive research field for both academia and industry. This review addresses the current finding on the biological activities of xanthene derivatives and discussed in detail some aspects of their structure-activity relationship (SAR).

Keywords: Xanthene, structure-activity relationship (SAR), biological activity, polycyclic, natural resources, pharmaceutical activities.

1. INTRODUCTION

Polycyclic aromatic hydrocarbons (PAHs) consist of fused benzene or hetero benzene rings in linear, angular or clustered arrangements [1]. PAHs are interested due to their chemical structures and biological activities [2-6]. Moreover, their toxicity and xenobiotic activities have been of interest since such chemical

the identification of novel xanthene derivatives with therapeutics properties.

2. ANTI-TUMOR ACTIVITIES

One of the most interest biological activities of xanthene core heterocyclic compounds is their anti-cancer activity. This feature

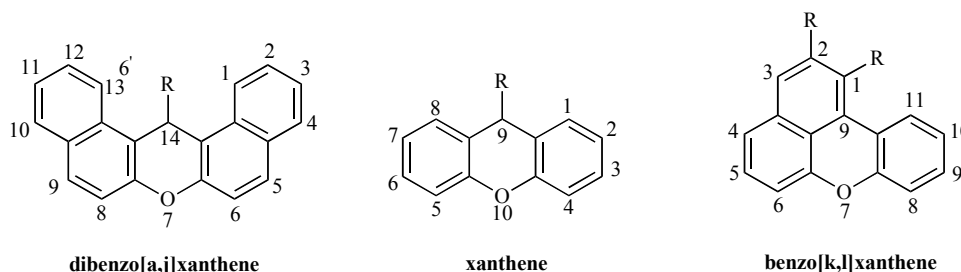


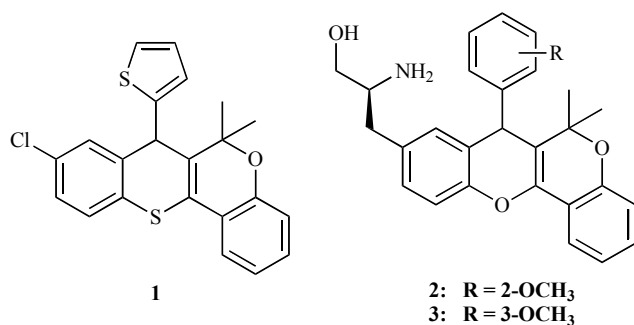
Fig. (1). Chemical structures of xanthene derivatives.

structures were found in pollutants [7-11]. Among all of the polycyclic compounds, xanthene derivatives have a very diverse application, for example in addition to their pharmaceutical application, these compounds have been used in laser technology [12-14], food industry [15-17], and fluorescent materials [18-24]. For example, some of the xanthene dyes such as rose bengal and phloxine B have been used as color additives in food industries [20, 25-27]. In our previous studies, we addressed xanthene scaffolds [28-32]. In this review, we aimed to address the biological activities of xanthene structures (Fig. 1) and discuss the structure-activity relationship (SAR) of xanthene scaffolds which might be useful for

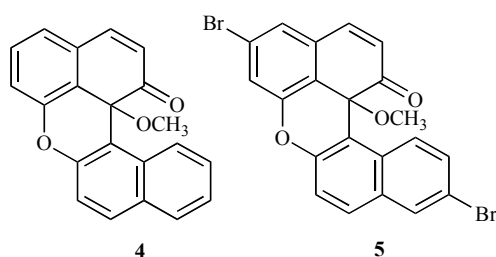
has been studied in several studies in which the important ones are discussed in the following section.

Kumar *et al.* synthesized a series of chromene-xanthene hybrids and studied their *in vitro* anticancer and antiproliferative properties on MCF-7 and MDA-MB-231, two human breast cancer cell lines [33]. Structure-activity studies showed that chromene-xanthene hybrids with phenol and thiophenol had lower anticancer activity profiles compared to that of hybrids with tyrosine moiety. For example, compound (1) with the thiophenol moiety displayed IC_{50} values of 22.3 μ M in MCF7 and 29 μ M against MDA-MB-231, while compound (2) was more active against MDA-MB-231 cell line (IC_{50} = 2.5 μ M); and compound (3) was the most active compound against MCF-7 (IC_{50} = 2.1 μ M). On the other hand, none of the compounds were toxic against non- cancer HEK-293 (human embryonic kidney) cell line [33].

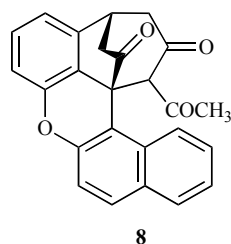
*Address correspondence to this author at the Department of Chemistry, Shahreza Branch, Islamic Azad University, Shahreza, Isfahan, 86145-311, Iran; Tel: 00983137927098; E-mail: nazifi@pharm.mui.ac.ir



In another study, Wang *et al.* synthesized four dibenzoxanthene compounds and evaluated their DNA interaction and cytotoxicity [34]. All of the tested compounds showed good growth inhibition against BEL-7402, Hela, MG-63, and SKBR-3 cell lines. Compound (5) showed the highest activity against BEL-7402 (IC_{50} = 10.1 μ M), Hela (IC_{50} = 9.4 μ M) and SKBR-3 (IC_{50} = 6.2 μ M) cell lines. Wang *et al.* reported that the introduction of the two bromine atoms in compound (4-7) structure enhanced its lipophilicity and cytotoxicity effects. Moreover, DNA-binding evaluation using electronic absorption titration and viscosity measurements revealed that the cytotoxicity of the tested molecules is not related to their tendency to bind DNA. Furthermore, assessment of the antioxidant activity of all of these compounds demonstrated that at a concentration of 3.5 mM these compound can effectively scavenge hydroxyl radical.



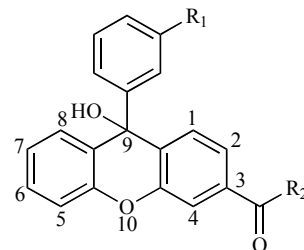
In another study by Wang *et al.* a novel ethanocycloheptono [3,4,5-kl]benzo[a]xanthene (8) was synthesized based on Michael's addition reaction and using dibenzoxanthene (4) as the parent compound [35]. The anticancer effects of this novel xanthene core compounds were studied using MTT assay against BEL-7402, HeLa, A549, and MG-63 cell lines which showed IC_{50} values of 17, 30, 54 and 82 μ M, respectively. BEL-7402 cell line was the most sensitive cell line to this compound. The cell death mechanism (apoptosis or necrosis) in this cell line was studied using AO/EB staining. It is shown that when BEL-7402 cells were exposed to compound (8), the intracellular ROS was elevated. This indicates a ROS-mediated apoptotic mechanism.



Giri *et al.* synthesized several di- and tri-substituted 9-alkyl xanthenols and tested their anti-cancer activities against Hela, DU-145, and MCF-7 cell lines [36]. Structure-activity relationship studies of these compounds suggested that introduction of a 3-methoxy group to the 9-phenyl ring (10; IC_{50} = 36 μ M vs. 11; IC_{50}

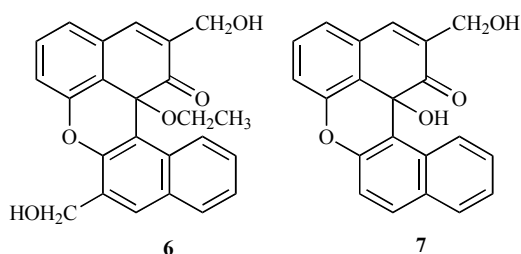
= 59 μ M; DU-145), and replacement of the diethylamine with the N-methylpiperazine (9; IC_{50} = 139 μ M vs. 10; IC_{50} = 36 μ M; DU-145), significantly enhanced their antitumor activity.

Among the tested compounds, compound (11) was the most potent one to inhibit cell proliferation with IC_{50} values ranging from 36-50 μ M. Based on the SAR analysis, the introduction of 7-fluoro substituent to (11) enhanced the anticancer activity with IC_{50} value lower than a micromolar range.



- 9: R₁ = OCH₃, R₂ = N-methylpiperazine
10: R₁ = H, R₂ = diethylamine
11: R₁ = OCH₃, R₂ = diethylamine

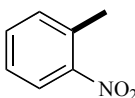
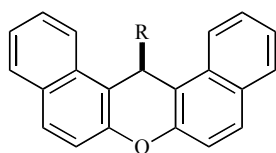
Bhattacharya (2009) reported the synthesis of several 14-aryl-14H-dibenzo[a,j]xanthene compounds and assessment of their antiproliferative activity against six human cancer cell lines (SW-620, Colo-205 and 502713 (Colon), SKNSH (CNS), A-549 (Lung) and PC-3 (Prostate)) [37]. Compound (16) was reported as the most



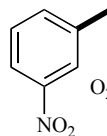
effective one against Colo-205 and Colon-502713 cell lines with IC_{50} values of 37.9 and 41.3 μ M, respectively. Structure-activity studies revealed that the aliphatic hydroxyl group in the para position of the aryl ring tends to increase the antiproliferative potency. Compound (15) which contains 4-hydroxy and 3-methoxy aryl groups also showed good cytotoxic activity against Colo-205 cell line with IC_{50} value of 41.9 μ M. Moreover, it was shown that substitution of the nitro group at the ortho position of the aryl ring (12) has elevated the antiproliferative potency, while *meta* and *para* substituted aryls (13 and 14 respectively) weaken the anticancer activity.

Spatofora *et al.* introduced a biomimetic synthesis of benzo[k,l]xanthene lignans and studied their DNA-interaction and anticancer activity against six cell lines including H226, A549 (human lung carcinoma), SH-SY5Y (human neuroblastoma), HCT-116, Caco-2, and HT-29 (human colon carcinoma) [38]. The antiproliferative test showed that except compound (22), other compounds were active against at least one cell line. Structure-activity relationship of (22) revealed that it has the lowest lipophilic properties, thus had the lowest activity toward the tested cell lines. On the other hand, compounds (18) and (21) showed the highest lipophilic properties and higher activity against HCT-116 cells compared to that of the positive control (5-FU) (22, GI_{50} = 4.22 μ M; 7, GI_{50} = 3.16 μ M; 5-FU, GI_{50} = 5.81 μ M).

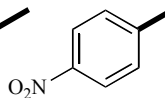
Micco *et al.* [39] performed a DNA-docking study on some of these compounds and reported that the hydroxy groups in C-6, C-9,



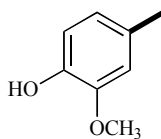
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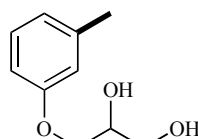
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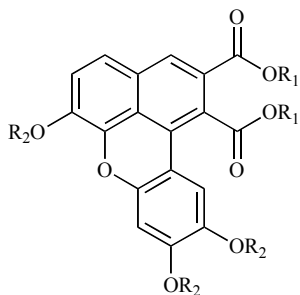


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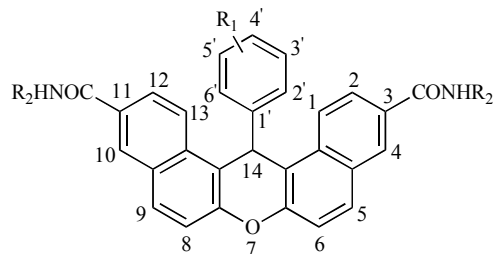
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and C-10 are essential for binding with guanines and anomeric oxygen of deoxyribose, while C-1/C-2 pendants were interacting with the grooves. These findings are in agreement with the anti-tumor activity results. For example, compared to compound (17), benzoxanthene (18) showed a higher affinity toward the minor groove and a higher anti-cancer activity against SW480 cells (GI_{50} : 4.76 and GI_{50} = 2.57 μ M, respectively). The methylated derivatives (19) and (20) were found to be inactive against SW480 cells.



- 17: $R_1 = CH_3, R_2 = H$
 18: $R_1 = CH_2CH_2Ph, R_2 = H$
 19: $R_1 = CH_3, R_2 = CH_3$
 20: $R_1 = CH_2CH_2Ph, R_2 = CH_3$
 21: $R_1 = R_2 = O(CH_2)_3CH_3$
 22: $R_1 = R_2 = OCH_2CH_2OH$

Song *et al.* synthesized and investigated the structure-activity relationship of several *N*-substituted 14-aryl-14*H* dibenzo[*a,j*]xanthene-3,11-dicarboxamides and their *in vitro* antitumor activity [40]. These compounds were tested against several human hepatoma cancer cell lines (SK-HEP-1, HepG2, SMMC-7721) and acute promyelocytic leukemia cells (NB4). According to the SAR analysis, different parameters such as side chains on C-3 and C-11 positions and the nature of the substituent at 14-phenyl group can significantly affect the anticancer activity. For example, carboxamide substituent at C-3 and C-11 position led to the production of more effective compounds. Moreover, the size of the substituent on the nitrogen atom of the carboxamide side-chain can lead to an effective inhibitory activity against the mentioned cell lines. In this regard, among (23a) to (26a) compounds, (23a) was the most active compound against all cell lines due to the small size of the positioned group on the nitrogen atom of the carboxamide side chain. The substituent at the *para* position of the 14-phenyl ring resulted in a higher inhibitory activity comparable to that of the *ortho* or *meta* positions. The impact of the ring substituent is reflected in the observed IC_{50} of (23c) and (23b) on HepG2 cells with IC_{50} of 9.68 μ M and >50 μ M, respectively. Moreover, halogen substituents at the *para* position of the 14-phenyl ring were more favorable than other substituents. The evaluated antitumor activity (IC_{50} values) against HepG2 for compounds (23f), (23g) and (23h) were 50, 50 and 39 μ M, respectively.



23: $R_2 = H$ 24: $R_2 = CH_3$

25: $R_2 = CH_2CH_2CH_3$

26: $R_2 = CH_2CH(CH_3)_2$

- a: $R_1 = H$ b: $R_1 = 2'-F$
 c: $R_1 = 4'-F$ d: $R_1 = 2'-Cl$
 e: $R_1 = 4'-Cl$ f: $R_1 = 3'-NO_2$
 g: $R_1 = 4'-NO_2$ h: $R_1 = 4'-CH_3$

3. ANTIBACTERIAL ACTIVITIES

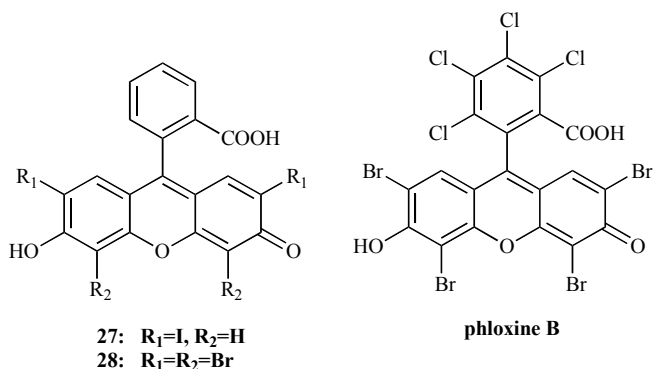
Another interesting biological activity of xanthene derivatives is their antimicrobial activity, in particular, antibacterial activity, which has been addressed in some studies.

Mihalyi *et al.* reported the antibacterial activities of some fluorescein derivatives as novel inhibitors of phospho-MurNAc-pentapeptide translocase MraY [41]. This enzyme and glycosyltransferase MurG are involved in the formation of lipid intermediates. Among fluorescein derivatives, phloxine B showed good inhibitory activity against MraY of *E. coli* with an *in vitro* IC_{50} value of 32 μ M. This compound also showed weaker IC_{50} values (128-254 μ M) against MraY enzymes of other bacteria and showed no activity against MurG protein. Phloxine B inhibits peptidoglycan synthesis of *Staphylococcus aureus* by inhibition of MraY through halogenation of the carbon skeleton. The inhibitory activity of diiodofluorescein (27) and tetrabromo derivative (28) (fluorescein derivatives) against MraY was higher than that of the MurG, (with IC_{50} values of 16 μ g/mL and 7 μ g/mL, respectively [42].

Subsequently, *in silico* simulation of binding of phloxine B to MraY of *Aquifex* indicated a good interaction with a GOLD score of -6.9 kcal/mol through the formation of two bonds between the carboxylic acid with the Mg^{2+} cofactor and Lys-121. The two aromatic rings can form favorable π -cation interaction with Lys-121 and Mg^{2+} cofactor which facilitated its fitting into the hydrophobic pocket of the active site.

4. ANTI-ALZHEIMER DISEASE

Alzheimer's is another target for the development of xanthene derivatives.



Epstein *et al.* synthesized several Aminooxazoline xanthenes as inhibitors of the β -site of the amyloid precursor protein cleaving enzyme 1 (BACE1) [43]. Compound (27) (BACE1 $IC_{50} = 2nM$), which was reported as a promising lead for the development of agents for Alzheimer's disease, demonstrated an inhibitory effect on hERG ion channel and also acted as a substrate for Pgp. Later they studied lowering the level of both hERG inhibition and Pgp efflux while maintaining the BACE1 inhibitory effect. SAR studies revealed that compound (30) (BACE1 $IC_{50} = 6 nM$) which is formed by substitution of 4-dihydropyranyl had a lower hERG inhibition activity while showing a desirable efflux ratio. For improving the inhibition activity against BACE1, in lead compound (29), a neopentyloxy substituent remained unchanged while compounds containing 2-fluoro-3-pyridinyl groups were synthesized. In this series, compound (31) with 2-fluoro-3-pyridinyl moiety showed an increased BACE1 inhibition ($IC_{50} = 0.8 nM$). In another study, compounds having a fluorine substitution at the position-4 displayed an improved inhibitory potency against

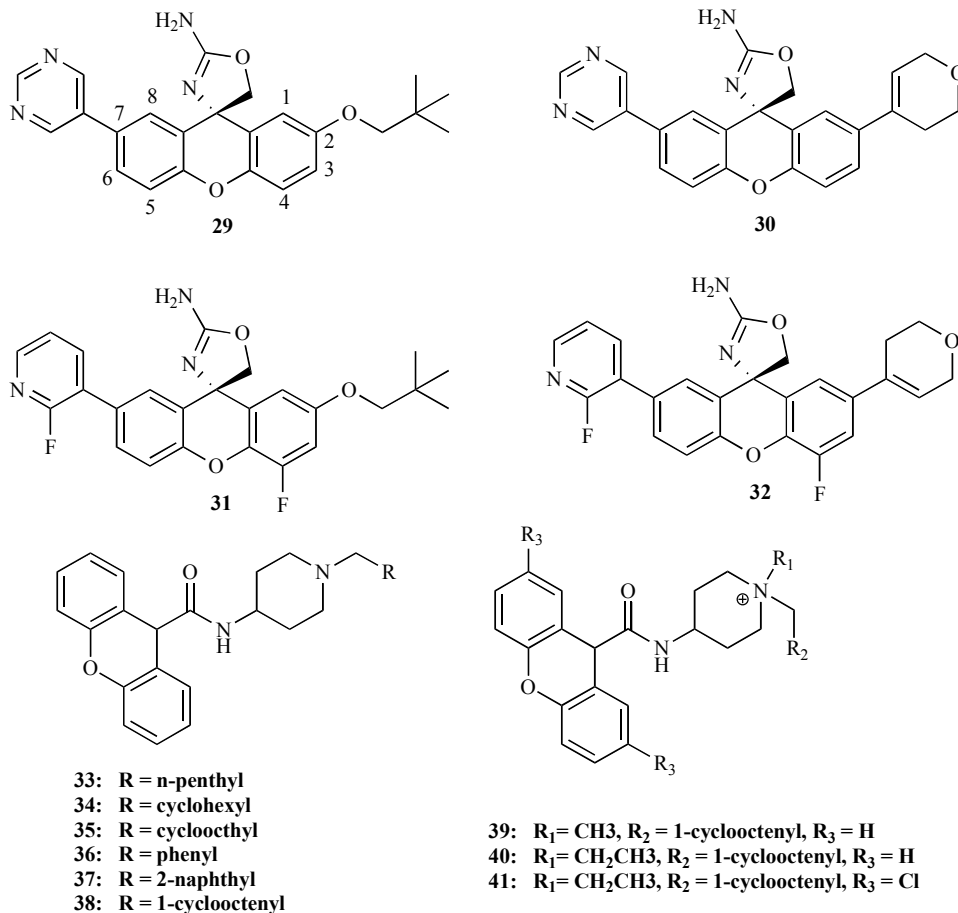
BACE1; on the other hand, substitution at the position-5 was not tolerated. Simultaneous substitutions of 4-fluoro and 2-fluoro-3-pyridyl on the xanthene moiety led to the discovery of compound (32) with excellent *in vitro* and *in vivo* profile.

5. ANTI-INFLAMMATORY ACTIVITIES

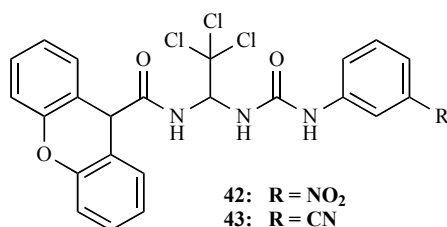
Xanthenes have been also tested for their anti-inflammatory activities. Naya *et al.* designed several human CCR1 receptor antagonists based on xanthenecarboxamide (33) [44]. This compound inhibited hCCR1 with IC_{50} value of 510 nM. The potent antagonists (comparable to that of the lead compound (33)) were screened through SAR analysis. According to the SAR studies, replacement of *n*-hexyl group on the piperidine ring with cyclohexyl (34), cyclooctyl (35) and 1-cyclooctenyl (38) groups enhanced the binding affinity toward human CCR1 receptors with IC_{50} values of 200, 140 and 51 nM, respectively. On the other hand, replacement with phenyl (36) and 2-naphthylmethyl (37) groups dramatically reduce the inhibitory effect ($IC_{50} = >10000 nM$). Moreover, quaternizing of the piperidine nitrogen with different alkyl groups resulted in higher inhibitory activity compared to that of the piperidine with the tertiary amine (39, $IC_{50} = 2.5nM$; 40, $IC_{50} = 2nM$). It is suggested that the piperidine nitrogen is bonded to anions by reversible ionic interactions to the binding pocket of hCCR1. Introduction of a chlorine atom at xanthene moiety position resulted in IC_{50} of 1.2 nM, indicating that compound (41) is a potent inhibitor of human CCR1 receptors.

6. ANTIDIABETIC ACTIVITIES

Diabetes is another target disease for xanthene compounds. Kwon *et al.* synthesized and tested two xanthene derivatives, 9H-xanthene-9-carboxylic acid [2,2,2-trichloro-1-[3-(3-nitro-phenyl)-



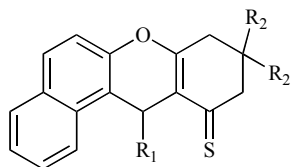
thioureido]-ethyl]-amide (42) and 9H-xanthene-9-carboxylic acid [2,2,2-trichloro-1-[3-(3-cyano-phenyl)-thioureido]-ethyl]-amide (43), for treatment of metabolic diseases such as diabetes mellitus [45]. The selection of xanthene core was due to the fact that it is the chemical backbone structure of mangiferin, a traditional drug for treating diabetes in Southeast Asia [46]. The AMPK activator properties (AMP-activated protein kinase) is tested when a research goal is the treatment of metabolic diseases such as diabetes mellitus. Both compounds, improved glucose homeostasis after administration of 3 mg/kg which was equal to the observed effects of 50 mg/kg of metformin *in vivo*. Moreover, (42) and (43) increased glucose tolerance, and also a single dose of 3 mg/kg increased phosphorylation of AMPK in skeletal muscle. Thus, (42) and (43) have higher treatment efficiency compared to that of metformin, hence, are potential candidates for the treatment of type 2 diabetes.



7. ANTIFUNGAL AND INSECTICIDAL ACTIVITIES

Xanthene compounds also tested for their fungicide and insecticidal activities. Khurana *et al.* produced several 12-aryl-8, 9, 10, 12-tetrahydrobenzo[a]xanthene-11-thiones and measured their urease inhibitory, antifungal (against *Rhizoctonia bataticola*, *Sclerotium rolfsii*, *Fusarium oxysporum*, and *Alternaria porii*) and insecticidal (against *Spodoptera litura*) activities [47]. All of the tested compounds showed moderate activity against all of the tested fungal strains. The highest activity was obtained against *Alternaria porii* with an ED₅₀ range of 135-670 µg/ml. Structure-activity studies showed that the introduction of 2-BrC₆H₄ (48) or 2-naphthyl (47) substituents on the tricyclic ring resulted in a good antifungal activity with ED₅₀ 165 and 135 µg/ml respectively.

The insecticidal activity of these compounds was tested against *Spodoptera litura*. The structural analysis suggested that 2-BrC₆H₄ (48) or 2-naphthyl (47) substituents increased insect mortality. Moreover, the introduction of 4-NO₂ substituent in the phenyl ring on the xanthene moiety increased the insect growth regulator activities with a 100% growth inhibition index.



- 44: R₁ = 4-ClC₆H₄, R₂ = CH₃
45: R₁ = 4-(CH₃)C₆H₄, R₂ = CH₃
46: R₁ = 3-(OCH₃), 4-(OH)C₆H₄, R₂ = CH₃
47: R₁ = 2-naphthyl, R₂ = CH₃
48: R₁ = 2-BrC₆H₄, R₂ = H
49: R₁ = C₆H₅, R₂ = H

Inhibition of urease activity was determined after 9 and 27 hours of incubation. Among the tested compounds, compound (49) showed the highest activity with 47 and 22% inhibition efficiency, respectively. SAR analysis suggests that substitution with electron-donating groups at the para position (44, 45) was the most successful one regarding urease inhibitory activity, however,

increasing the size of the aryl unit (46, 47) reduced urease inhibition.

CONCLUSION

This review reported the different biological activities of xanthene based derivatives which are potential chemicals for the treatment of infectious diseases (such as bacterial and fungal), tumor proliferation, Alzheimer's diseases, diabetic disorder and inflammation with hCCR1 receptor blocking.

Considering the critical issue of the superbugs (the infectious pathogens which developed resistance toward drugs), identifying new structural classes such as xanthenes is a must. As an antimicrobial agent, these compounds are broad-spectrum antibacterial agents with activity against both gram negative and positive bacteria and showed antifungal activity against several strains.

SAR analysis of the antitumor activity of these chemicals revealed that the phenyl ring on the C-14 position with a substituent at the para position enhanced anti-cancer activity. Moreover, the lipophilicity level can also affect the chemotherapeutic properties. Considering the diverse range of antitumor activities displayed by these compounds, identification of new antitumor drug candidates from this class of compounds or even enhancement of the already existing antitumor chemicals by adding this structure is a potential development application for this class of chemicals.

In terms of anti- Alzheimer's activity, the presence of aminooxazoline moiety on C-14 position is necessary for the inhibitory activity of such compounds against BACE1. Moreover, fluorine substitution in the position-4 of the xanthene core increased the BACE1 inhibitory activity.

In conclusion, xanthene not only is an industrially important compound but due to having various biological activities, it has great potential to be used as a core chemical structure for the development of novel therapeutic agents against many types of diseases.

CONSENT FOR PUBLICATION

Not applicable.

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None.

CONFLICT OF INTEREST

The authors declare no conflict of interest, financial or otherwise.

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Declared none.

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